

Erin Kwon

Berkeley, CA | gwonehleen@berkeley.edu | 510-914-0416 | [LinkedIn](#) | [Portfolio](#)

EDUCATION

University of California, Berkeley

B.S. in Mechanical Engineering

Expected May 2027

GPA: 3.7/4.0

Relevant Coursework: 3D Modeling and Design, Computer Programming, Manufacturing and Design Communication, Electronics for the IoT, Thermodynamics, Solid Mechanics, Statistics & Data Science

EXPERIENCE

Product Development Engineering Intern | Jupiter Endovascular – Menlo Park, CA *June 2025 - Aug 2025*

- Prototyped and validated catheter handles and shafts by iterating on a force, leakage, and aging test matrix
- Developed and verified Gen 1.5 Handle test methods using Zwick tensile testing and MATLAB data analysis
- Led Gen 1.5 Handle press fit design through prototyping, ordering samples, tensile tests, and analyzing results
- Molded and evaluated 5 medical seal types using Keyence for dimensional testing and compression analysis
- Documented and presented test methods, data, and results to cross-functional teams, driving design iterations

R&D Engineering Intern | MUNE – Seoul, South Korea

Sept 2025 - Nov 2025

- Performed occlusion, transducer, and flow sensor testing across multiple materials and motor configurations
- Conducted volume testing for Regurge and Flush stages, identifying leakage and flow inconsistencies
- Optimized test reliability by troubleshooting variables such as tubing friction, pump gap, and IV bag pressure
- Supported alarm functionality testing with heart monitoring equipment to validate system safety thresholds
- Created data collection documentation and organized large datasets to support review and validation cycles

Undergraduate Research Assistant | Berkeley Sensor & Actuator Center

Mar 2025 - Jan 2026

Liwei Lin Lab: Research on wearable ultrasound imaging devices using pMUTs for neuromodulation applications

- Designed a custom PCB with KiCad to support ESP32-based testing for ultrasound experiments
- Built a Schlieren imaging setup to visualize acoustic waves generated by pMUTs and fluid flow disturbances
- Contributing to the development of pMUT technology for potential commercialization in a startup

Exoskeleton Research Assistant | University of California, Berkeley

Feb 2024 - May 2025

Nanotechnology Unleashed: Investigation of full-body exoskeleton design and nanostructures for energy storage

- Developed a wrist exoskeleton for stroke rehabilitation patients to assist in effective motor function recovery
 - Designed a screw-lock syringe holder for microfluidics, reducing fluid transfer time by 25%
 - Created a 3D model of an adapter in SolidWorks to connect microfluidic channels for nanotubes
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PROJECTS

EnableTech | SolidWorks, Fusion 360, Markforged X7, Tormach CNC

Sep 2024 - May 2025

- Designed a mobile platform to assist a little person in safely transitioning into and out of a wheelchair
- Generated G-code using CAM software to CNC custom mounts for secure wheelchair integration
- Iterated through 5+ stages of rapid prototyping, utilizing carbon fiber printing and machine shop tools

Smart Golf Club | ESP32, MicroPython, Sensors, Actuators

Feb 2025 - May 2025

- Built a smart golf club with instantaneous haptic, visual, and auditory feedback to improve swing technique
 - Integrated motion, pressure, and resistive sensors on the club and a wrist-worn device to track swing metrics
 - Calibrated thresholds for grip pressure, swing tempo, and clubface alignment through iterative testing
 - Enabled wireless feedback via synchronized ESP32 communication and real-time data display in Thonny
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SKILLS

Technical Skills: SolidWorks, Fusion 360, Onshape, AutoCAD, Python, MATLAB, KiCad

Test Method Development: ASTM-aligned testing, custom STM fixture design, data collection & analysis, performed automated visual inspections using Keyence systems

Prototyping & Manufacturing: Rapid prototyping, fixture fabrication, part integration, machining